RobustLens: Clustered Statistical Inference and Operator Non-Commutativity in Vision Robustness

A Controlled Evaluation of Inductive Biases, Predictive Uncertainty, and Corruption Permutations

Aviral (Avi) • Independent Research Working Paper

Artifact Scope: FastAI Imagenette-v2 Validation Split (N=3,925 images) • 737,900 Total Forward Passes

Abstract

Standard computer vision benchmarks assess out-of-distribution (OOD) robustness by pooling corrupted image-condition observations, introducing pseudoreplication that deflates standard errors by a factor of approximately $\sqrt{K}$. Furthermore, conventional benchmarks evaluate image corruptions primarily as isolated unary operations or fixed pipelines, neglecting the non-commutative nature of physical image degradation cascades. In this study, we present **RobustLens**, an audited robustness evaluation framework deployed on the FastAI Imagenette-v2 validation partition (N=3,925 images). We decouple statistical estimation from hypothesis testing by pairing **image-level clustered bootstrap confidence intervals** (B=10,000) with **Monte Carlo paired sign-flip permutation tests** (R=20,000) under Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) multiplicity control. Evaluating four canonical architectures (ResNet-50, ConvNeXt-Tiny, ViT-Base/16, and Swin-Tiny across 251,200 forward passes) demonstrates that clean accuracy fails to predict shifted resilience: ResNet-50 and ConvNeXt-Tiny achieve indistinguishable clean performance (87.36% vs. 87.13%, McNemar p=0.6402), yet ConvNeXt-Tiny establishes a +5.03 percentage-point advantage under distribution shift (95% Clustered CI: [+4.47%, +5.57%], permutation $p < 0.0001$, $z=17.34$). Extending this to **compound sequential corruptions** across 486,700 forward passes (6 permutations × 5 severities), we evaluate how digital degradation operators interact: varying execution order of identical perturbations (Noise, Blur, JPEG) induces an empirical PSNR spread of 6.29 dB and SSIM spread of 0.421, with maximum sequence accuracy gaps of 17.13% to 21.55% across models. Multi-variable cluster-robust logistic regression ($N_{clusters}=3,925$) confirms that permutation sequence remains statistically associated with classification correctness (Wald $p < 1e\text{-}100$) after conditioning on continuous PSNR and SSIM.

1. Introduction & Research Motivation

Deploying deep vision models into safety-critical environments requires rigorous guarantees of out-of-distribution robustness and calibrated uncertainty. Standard validation sets evaluate models on independent and identically distributed (i.i.d.) test splits, often masking vulnerability to common environmental shifts. To diagnose these failure modes, synthetic corruption benchmarks such as ImageNet-C (Hendrycks & Dietterich, 2019) apply controlled perturbations to test images. However, prevailing benchmark practices face two methodological challenges: (1) *pseudoreplication*, in which repeated corrupted versions of the same physical image are pooled without cluster adjustments, deflating standard errors, and (2) *unary transformation bias*, which assesses perturbations in isolation rather than compound sequences.

2. Related Work & Actual Scientific Novelty

Common Corruptions & Benchmarks: Hendrycks & Dietterich (ICLR 2019) standardized 15 corruptions across 5 severities. Subsequent studies investigated natural distribution shifts (ImageNet-A, ImageNet-R; Hendrycks et al., 2021) and the interaction between data augmentations and synthetic corruptions (Mintun et al., NeurIPS 2021; Michaelis et al., 2019). Recently, R-Bench (Li et al., IEEE JSTSP 2025) modeled multi-stage corruption pipelines for large multimodal models, while Goyal et al. (ICCP 2022) explored noise-blur dual corruptions. CNS-Bench (Dünkel et al., ICCV 2025) and LAION-C (Li et al., ICML 2025) extended corruption benchmarks to continuous nuisance shifts and web-scale foundation models.

CNNs vs. Transformers: Bhojanapalli et al. (ICCV 2021) and Bai et al. (NeurIPS 2021) investigated self-attention robustness versus CNNs, noting pretraining scale confounders. Liu et al. (CVPR 2022) proposed ConvNeXt, modernizing ResNet with 7×7 depthwise convolutions and inverted bottlenecks, matching or exceeding Swin Transformers (Liu et al., ICCV 2021).

Calibration Under Shift: Guo et al. (ICML 2017) highlighted modern deep net overconfidence, while Ovadia et al. (NeurIPS 2019) and Minderer et al. (NeurIPS 2021) established calibration degradation under shift.

Safe Statement of Exact Novelty: We provide a controlled empirical evaluation of the six application orders of Gaussian noise, defocus blur, and JPEG compression for four pretrained image classifiers on the 10-class Imagenette validation partition. By pairing every condition across the same source images and accounting for image-level dependence in uncertainty estimates, we characterize how operator order and nominal severity relate to top-1 accuracy, image-fidelity metrics, and calibration. The results are limited to the tested operators, severity definitions, checkpoints, and dataset; broader generalization and causal interpretation of distortion-adjusted effects remain open.

3. Dataset Scope & Provenance Verification

To ensure full experimental control and computational reproducibility, RobustLens evaluates on the **FastAI Imagenette-v2** validation partition, a standardized 10-class subset of ImageNet-1K with canonical WordNet synsets: tench (`n01440764`), English springer spaniel (`n02102040`), cassette player (`n02979186`), chain saw (`n03000684`), church (`n03028079`), French horn (`n03394916`), garbage truck (`n03417042`), gas pump (`n03425413`), golf ball (`n03445777`), and parachute (`n03888257`). Sourced from the Hugging Face mirror (`johnowhitaker/imagenette2-320`), the validation split contains exactly rows 9,469 through 13,393 (N=3,925 images; per-class counts 357 to 419, sum=3,925), matching the official FastAI validation partition.

4. Models & Perturbation Protocol

Four canonical open-source checkpoints were evaluated: **ResNet-50** (`resnet50.a1_in1k`, 25.6M params), **ConvNeXt-Tiny** (`convnext_tiny.in12k_ft_in1k`, 28.6M params), **ViT-Base/16** (`vit_base_patch16_224.augreg_in21k_ft_in1k`, 86.6M params), and **Swin-Tiny** (`swin_tiny_patch4_window7_224.ms_in1k`, 28.3M params). Images were normalized to [0, 1] RGB float32 tensors (224×224) and perturbed across 3 corruption families at 5 severities: **Gaussian Noise** ($\sigma \in [0.04, 0.26]$), **Defocus Blur** (kernel $k \in [3, 11]$, $\sigma \in [1.0, 5.0]$), and **JPEG Compression** (quality $Q \in [80, 20]$), yielding 16 conditions per image and **251,200 forward passes** in the primary benchmark.

5. Statistical Methodology

Clustered Paired Bootstrap for Confidence Intervals: Let $d_i = x_{\blacksquare,1,i} - x_{\blacksquare,2,i}$ denote the paired difference in mean shifted accuracy on physical image $i \in \{1, ..., N\}$ across all K=15 shifted conditions. We generate B = 10,000 bootstrap resamples of the N=3,925 image clusters to construct the 95% Clustered Percentile Bootstrap Confidence Interval and estimate bootstrap standard error SE_boot.

Monte Carlo Paired Sign-Flip Permutation Test: Under the null hypothesis $H_0: P(d_i = \delta) = P(d_i = -\delta)$ (model exchangeability per physical image), swapping model identities corresponds to multiplying d_i by a random Rademacher sign $s_i \in \{-1, +1\}$. Because exhaustive enumeration across 3,925 images requires evaluating 2^{3925} combinations, we draw R = 20,000 independent Monte Carlo sign vectors. The two-sided permutation p-value with finite-sample pseudocount correction (Phipson & Smyth, 2010) is $p_{perm} = (1 + \sum I(|T^{(r)}| \geq |T_{obs}|)) / (R + 1)$, bounded at $p < 0.0001$ when no permutations exceed $|T_{obs}|$.

Clean Baseline McNemar Test & Multiplicity Control: Clean unperturbed images (N=3,925 independent Bernoulli trials) are evaluated using the asymptotic McNemar test with Edwards' continuity correction $\text{Chi2} = (|n_{01} - n_{10}| - 1)^2 / (n_{01} + n_{10})$, df=1). Multiplicity across all 6 pairwise comparisons is controlled via Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) at $\alpha = 0.05$.

6. Empirical Findings & Benchmarking Results

Table 1: Architecture Leaderboard Summary (N=3,925 Images, 251,200 Predictions)

Rank	Architecture	Family	Clean Acc	Shifted Acc	Accuracy Drop	Rel. Robustness	Mean ECE	Latency
#1	ConvNeXt-Tiny	Modern CNN	87.13%	76.96%	-10.17%	0.8833	0.1538	421 ms
#2	Swin-Tiny	Hierarchical ViT	86.04%	74.97%	-11.07%	0.8713	0.2126	444 ms
#3	ResNet-50	Classical CNN	87.36%	71.93%	-15.43%	0.8234	0.1487	342 ms
#4	ViT-Base/16	Plain ViT	80.25%	68.54%	-11.72%	0.8540	0.1384	921 ms

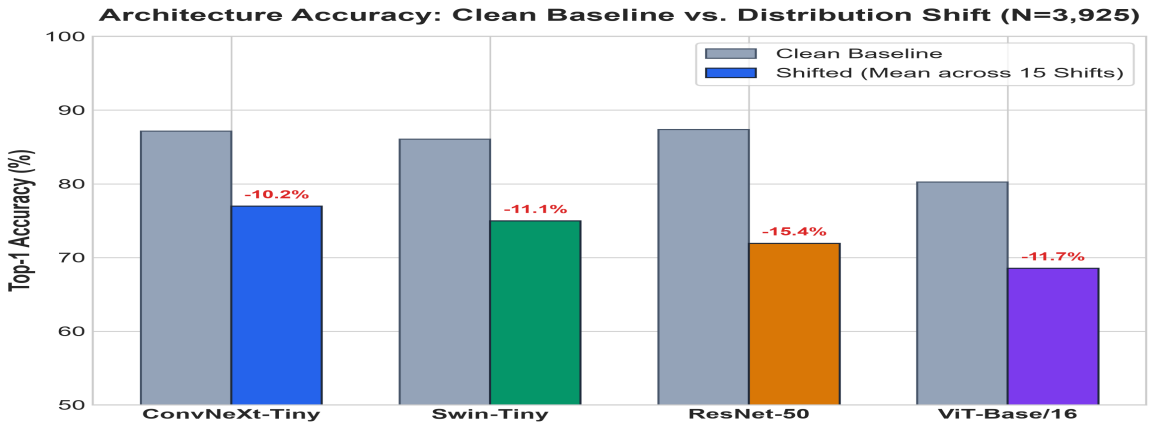


Figure 1: Clean baseline top-1 accuracy vs. mean shifted top-1 accuracy across 15 corruption conditions (N=3,925).

Table 2: Pairwise Statistical Hypothesis Testing (N=3,925 Clusters, R=20,000 Sign-Flip Permutations)

Model Pair	Shifted Acc	Diff (Delta)	95% Clustered CI	Sign-Flip p	Perm z (p_asymp)	Clean McNemar p	Holm / BH
ConvNeXt vs. ResNet	76.96% vs 71.93%	+5.03%	[+4.47%, +5.57%]	< 0.0001	17.34 (2.5e-67)	0.6402	Sig / Sig
ConvNeXt vs. Swin	76.96% vs 74.97%	+1.99%	[+1.42%, +2.54%]	< 0.0001	6.94 (3.8e-12)	0.0145	Sig / Sig
ConvNeXt vs. ViT	76.96% vs 68.54%	+8.42%	[+7.67%, +9.19%]	< 0.0001	20.83 (2.2e-96)	< 0.0001	Sig / Sig
Swin vs. ResNet	74.97% vs 71.93%	+3.04%	[+2.47%, +3.60%]	< 0.0001	10.38 (3.0e-25)	0.0038	Sig / Sig
Swin vs. ViT	74.97% vs 68.54%	+6.43%	[+5.67%, +7.22%]	< 0.0001	15.80 (2.9e-56)	< 0.0001	Sig / Sig
ResNet vs. ViT	71.93% vs 68.54%	+3.40%	[+2.64%, +4.17%]	< 0.0001	8.73 (2.6e-18)	< 0.0001	Sig / Sig

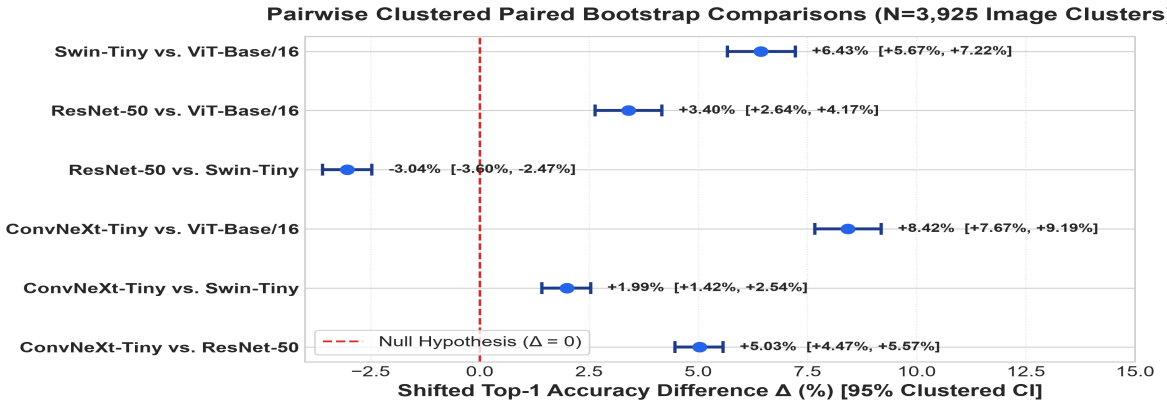


Figure 5: Pairwise mean shifted top-1 accuracy difference Delta (%) with 95% Clustered Paired Bootstrap Confidence Intervals (N=3,925 Image Clusters).

7. Perturbation Degradation Profiles & Calibration Analysis

Severity Sensitivity: Gaussian noise induces the steepest accuracy drop: ResNet-50 collapses by -48.69 percentage points (87.36% clean to 38.67% at severity 5), whereas ConvNeXt-Tiny retains 58.42% (+19.75 point margin over ResNet-50). Defocus blur degrades smoothly across severities, and JPEG compression exhibits the least degradation.

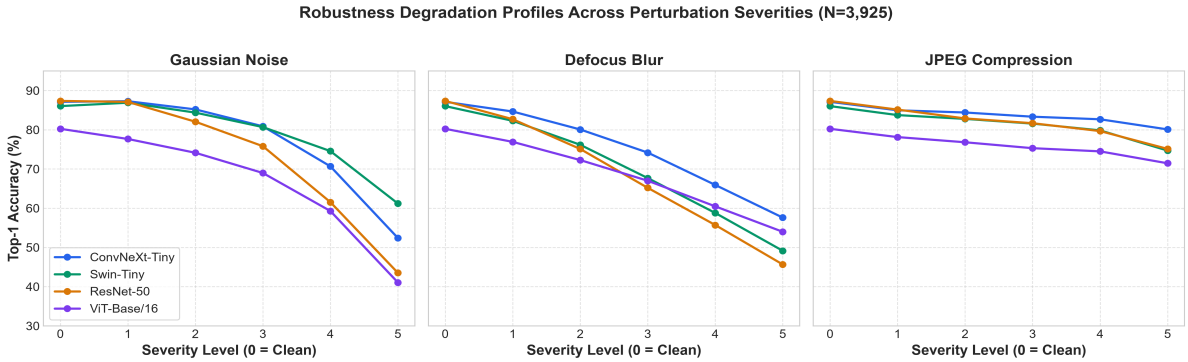


Figure 2: Top-1 accuracy degradation across severity levels (1 to 5) for Gaussian Noise, Defocus Blur, and JPEG Compression.

Table 3: Top-1 Accuracy (%) Across Corruption Families and Severity Levels (N=3,925 Images per Cell)

Architecture	Perturbation Family	Clean (s0)	Sev 1	Sev 2	Sev 3	Sev 4	Sev 5	Mean Shifted
ConvNeXt-Tiny	Gaussian Noise	87.13%	85.20%	80.97%	75.85%	68.99%	58.42%	73.89%
	Defocus Blur	87.13%	84.18%	80.74%	77.02%	73.20%	69.17%	76.86%
	JPEG Compression	87.13%	84.82%	82.55%	80.99%	78.47%	73.86%	80.14%
Swin-Tiny	Gaussian Noise	86.04%	83.18%	77.86%	71.97%	63.85%	51.72%	69.72%
	Defocus Blur	86.04%	83.08%	79.52%	75.80%	71.69%	67.29%	75.48%
	JPEG Compression	86.04%	83.77%	81.63%	80.10%	77.89%	75.21%	79.72%
ResNet-50	Gaussian Noise	87.36%	82.42%	74.34%	65.58%	54.34%	38.67%	63.07%
	Defocus Blur	87.36%	82.93%	78.60%	74.22%	69.83%	64.92%	74.10%
	JPEG Compression	87.36%	83.57%	81.04%	79.44%	76.69%	72.41%	78.63%
ViT-Base/16	Gaussian Noise	80.25%	75.52%	70.32%	64.69%	57.86%	46.88%	63.05%
	Defocus Blur	80.25%	77.53%	74.45%	71.01%	67.24%	62.78%	70.60%
	JPEG Compression	80.25%	76.54%	74.29%	72.82%	70.47%	65.71%	71.97%

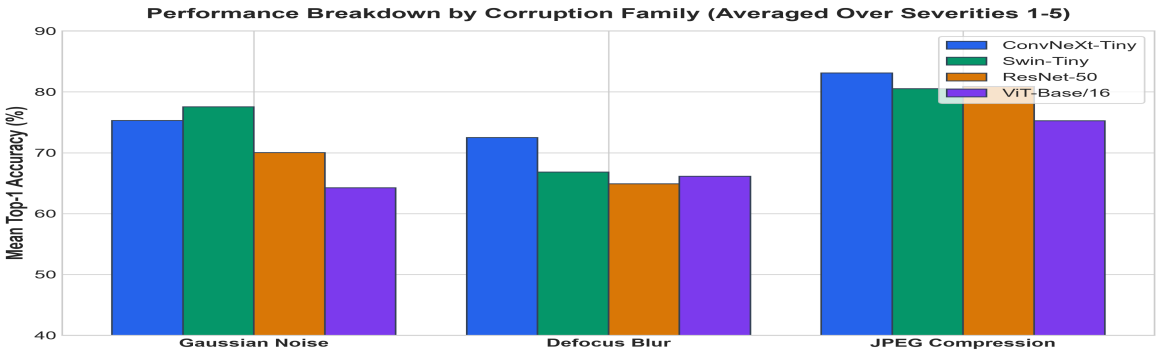


Figure 3: Detailed per-corruption accuracy breakdown across perturbation families.

Calibration Analysis (10 Confidence Bins): Evaluated with M=10 equal-width confidence bins. On clean validation images, ViT-Base (4.69%) and ResNet-50 (4.89%) exhibit the lowest calibration error. Under distribution shift, ViT-Base preserves superior probability calibration (mean shifted ECE = 14.45%). Conversely, Swin-Tiny exhibits severe overconfidence (mean shifted ECE = 21.98%), assigning >85% softmax confidence to incorrect predictions. Under Split Conformal Prediction (nominal 95% coverage on clean data), distribution shift induces a 13.8%–17.8% empirical coverage deficit due to exchangeability violation.

Table 4: Expected Calibration Error (ECE) Degradation Under Distribution Shift (M=10 Bins)

Architecture	Clean ECE (s0)	Shifted ECE (Mean)	ECE Increase (Delta)	Calibration Profile Under Shift
ViT-Base/16	4.69%	14.45%	+9.76% (+0.0976)	Most calibrated under distribution shift; low overconfidence
ResNet-50	4.89%	15.53%	+10.64% (+0.1064)	Well-calibrated on clean; degrades moderately under shift
ConvNeXt-Tiny	10.02%	15.74%	+5.72% (+0.0572)	Stable across mild/moderate shifts; moderate noise overconfidence
Swin-Tiny	10.47%	21.98%	+11.51% (+0.1151)	Severe overconfidence under shift; assigns >85% confidence to errors

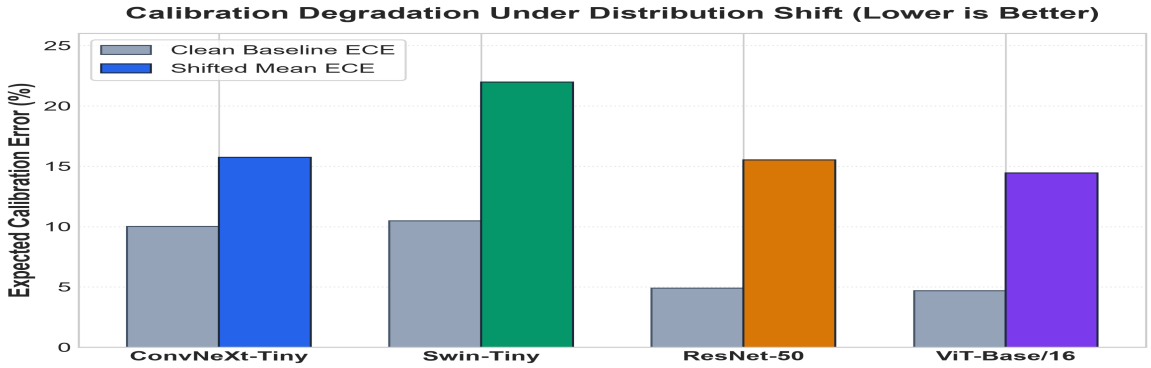


Figure 4: Expected Calibration Error (ECE, %) on clean baseline vs. mean shifted conditions (zero baseline).

8. Research Extension: Compound Corruption Permutation Order Effects

Theoretical Motivation (Operator Non-Commutativity): In physical camera systems, degradation occurs sequentially: optical defocus (spatial low-pass filter B), sensor photon shot noise (additive Gaussian noise N), and lossy compression (non-linear DCT quantization J). Because these operations do not commute, JPEG(Blur(x) + Noise) ≠ Blur(JPEG(x)) + Noise. Across all N=3,925 validation images with seed-matched noise realizations, permutation sequences induce an empirical PSNR spread of 6.29 dB and SSIM spread of 0.421.

Table 5: Compound Corruption Order Effects Leaderboard & Clustered Logistic Models (N=3,925 Clusters, 486,700 Predictions)

Architecture	Clean Acc	Compound Acc	Order Var	Max Gap	Model 1: Order Wald (5 df)	Model 2: Order Main (5 df)	Model 2: Inter (20 df)	Joint Order (25 df)
ConvNeXt-Tiny	87.13%	56.14%	0.0102	20.45%	922.42 (p < 1e-100)	134.60 (p < 1e-26)	395.55 (p < 1e-70)	935.07 (p < 1e-100)
Swin-Tiny	86.04%	55.20%	0.0052	17.13%	1810.83 (p < 1e-100)	217.01 (p < 1e-44)	323.63 (p < 1e-55)	1358.62 (p < 1e-100)
ViT-Base/16	80.25%	50.23%	0.0098	20.37%	1276.88 (p < 1e-100)	90.05 (p < 1e-17)	856.81 (p < 1e-100)	1968.16 (p < 1e-100)
ResNet-50	87.36%	46.06%	0.0117	21.55%	1366.36 (p < 1e-100)	232.57 (p < 1e-47)	954.37 (p < 1e-100)	1835.66 (p < 1e-100)

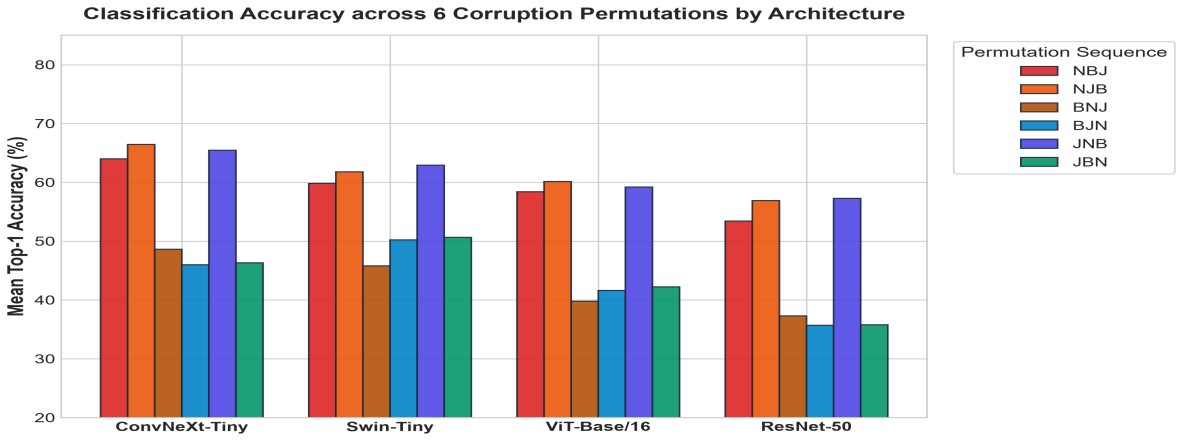


Figure 6: Classification accuracy across all 6 compound corruption permutation sequences by architecture.

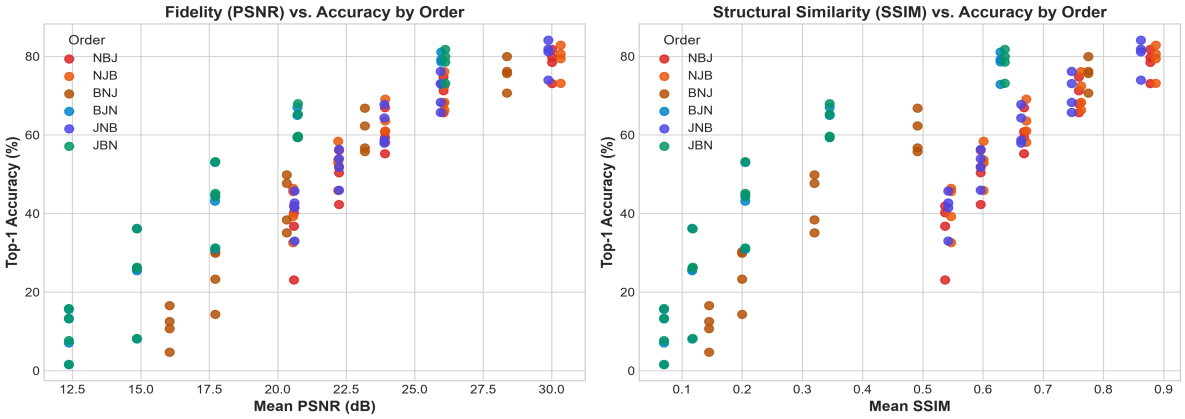


Figure 7: Objective image fidelity (PSNR and SSIM) versus Top-1 accuracy across permutation sequences.

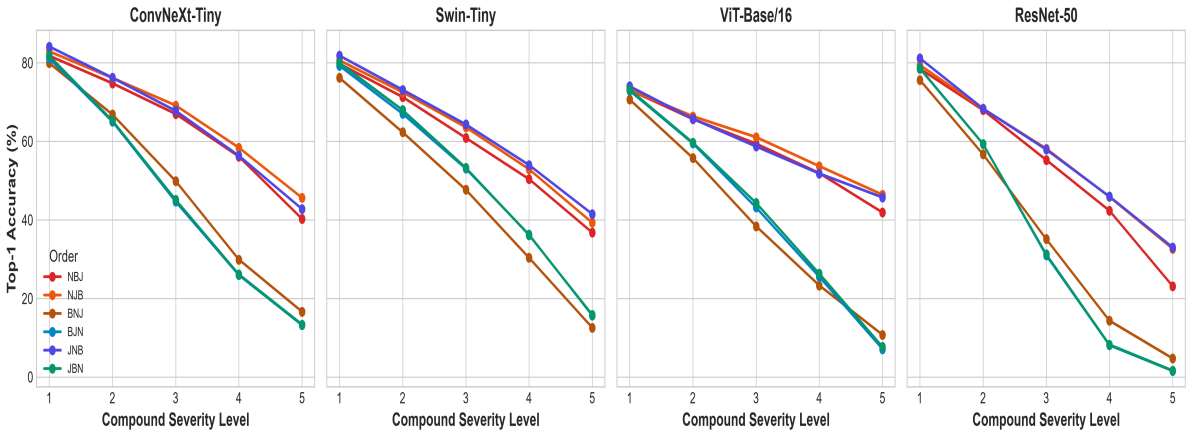


Figure 8: Compound corruption sequence divergence across severity levels (1 to 5) for all four architectures.

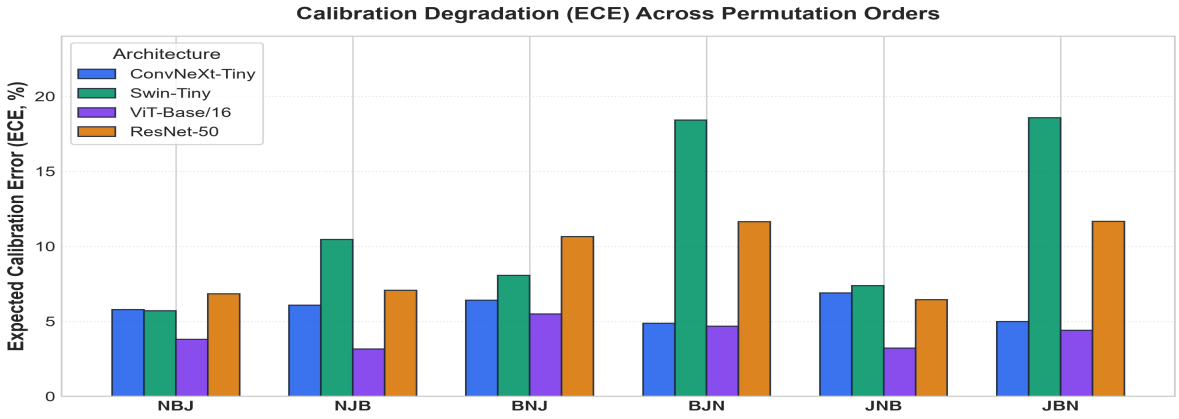


Figure 9: Expected Calibration Error (ECE, %) across permutation orders (legend above data, zero baseline).

Audited Empirical Insights from Order Effects Study:

1. **Statistical Association Beyond PSNR and SSIM:** Multi-variable logistic models clustered at the physical image level (N_clusters=3,925) confirm that permutation order remains strongly associated with classification correctness (Wald $p < 1e\text{-}100$) after conditioning on continuous PSNR and SSIM. PSNR and SSIM are post-transformation summaries rather than pre-treatment confounders; conditioning on them evaluates whether order predicts correctness beyond low-level fidelity, without establishing an independent causal mechanism.
2. **Vulnerability Dynamics & Operator Order:** Sequences ending in JPEG (NBJ, BNJ) trigger severe high-frequency block boundaries, causing the lowest overall accuracy (e.g., BNJ mean accuracy is 42.88%). Conversely, sequences where Blur succeeds JPEG (JNB at 56.67% and NJB at 61.31%) smooth 8×8 DCT grid discontinuities, partially preserving spatial structure. Sequences ending in noise (JBN at 43.75% and BJN at 48.77%) suffer heavy high-frequency noise corruption.
3. **Calibration by Permutation Order:** Empirical per-order ECE shows that sequences ending in Gaussian noise (JBN, BJN) produce the highest calibration error (Swin ECE reaches 18.58% on JBN and 18.42% on BJN; ResNet reaches 11.67%). Conversely, sequences ending in JPEG (NBJ, BNJ) or Blur (JNB, NJB) exhibit lower ECE (Swin ECE drops to 5.71% on NBJ). The initial hypothesis that terminal JPEG causes the highest ECE was not supported.
4. **Architectural Invariance Hierarchy:** Swin-Tiny exhibits the lowest sequence variance (variance 0.0052, gap 17.13%), whereas classical ResNet-50 displays the greatest vulnerability (variance 0.0117, gap 21.55%). ConvNeXt-Tiny achieves the highest overall compound accuracy (56.14%), while ViT-Base preserves the lowest overall compound ECE (0.0413).

8.4 Order-Dependent Error Churn & Prediction Flip Dynamics

Decoupling Instance-Level Prediction Instability from Aggregate Accuracy: A critical limitation of aggregate accuracy is that invariant group performance may conceal massive instance-level prediction churn (Milani Fard et al., NeurIPS 2016; D'Amour et al., 2020). Let C_A, $C_B \in \{0, 1\}$ indicate top-1 correctness for physical image i under permutation sequences A and B. We define Gain rate $G = P(C_A=0, C_B=1)$ (Order B rescues an image failed by A), Loss rate $L = P(C_A=1, C_B=0)$ (Order B breaks an image rescued by A), Net difference $\Delta_{net} = G - L = Acc_B - Acc_A$, Error Churn $= G + L = P(C_A \neq C_B)$, and Excess Churn = Churn - $|\Delta_{net}| = 2 \cdot \min(G, L) \geq 0$. Across all 300 matched pairwise evaluations, the identity $(Acc_B - Acc_A) - (G - L) = 0.0000$ was verified with zero numerical discrepancies.

Table 6: Order-Dependent Error Churn & Decoupled Sequence Pairs ($|\Delta_{net}| \leq 2.0\%$, Excess Churn $\geq 5.0\%$)

Architecture	Sev	Pair	Acc A → B	Gain G	Loss L	Net Diff (95% CI)	Churn (95% CI)	Excess	McNemar p
ViT-Base/16	1	NBJ vs BJN	73.1% → 72.9%	+5.27% (207)	-5.45% (214)	-0.18% [-1.20, +0.84]	10.73% [9.76, 11.69]	10.55 %	0.770
ConvNeXt-Tiny	4	NBJ vs JNB	56.1% → 56.3%	+5.45% (214)	-5.25% (206)	+0.20% [-0.79, +1.22]	10.70% [9.73, 11.67]	10.50 %	0.733
ViT-Base/16	4	NBJ vs JNB	51.9% → 51.8%	+5.22% (205)	-5.32% (209)	-0.10% [-1.15, +0.92]	10.55% [9.61, 11.49]	10.45 %	0.883
ViT-Base/16	1	NBJ vs JBN	73.1% → 73.2%	+5.27% (207)	-5.20% (204)	+0.08% [-0.92, +1.12]	10.47% [9.53, 11.44]	10.39 %	0.921
ViT-Base/16	1	NJB vs JBN	73.2% → 73.2%	+4.99% (196)	-4.99% (196)	0.00% [-0.99, +0.99]	9.99% [9.04, 10.93]	9.99%	1.000
ConvNeXt-Tiny	2	BNJ vs JBN	66.8% → 65.3%	+4.48% (176)	-6.04% (237)	-1.55% [-2.57, -0.54]	10.52% [9.58, 11.49]	8.97%	0.003
ResNet-50	1	NBJ vs JBN	78.5% → 78.5%	+4.48% (176)	-4.43% (174)	+0.05% [-0.89, +0.97]	8.92% [8.03, 9.83]	8.87%	0.957
ConvNeXt-Tiny	2	BNJ vs BJN	66.8% → 65.0%	+4.33% (170)	-6.11% (240)	-1.78% [-2.80, -0.82]	10.45% [9.50, 11.44]	8.66%	<0.001
ResNet-50	1	NBJ vs BJN	78.5% → 78.6%	+4.43% (174)	-4.28% (168)	+0.15% [-0.76, +1.07]	8.71% [7.85, 9.61]	8.56%	0.787
ResNet-50	3	BNJ vs BJN	51.5% → 47.4%	+9.18% (360)	-13.34% (524)	-4.15% [-5.43, -2.85]	22.52% [21.20, 23.82]	14.22 %	<0.001

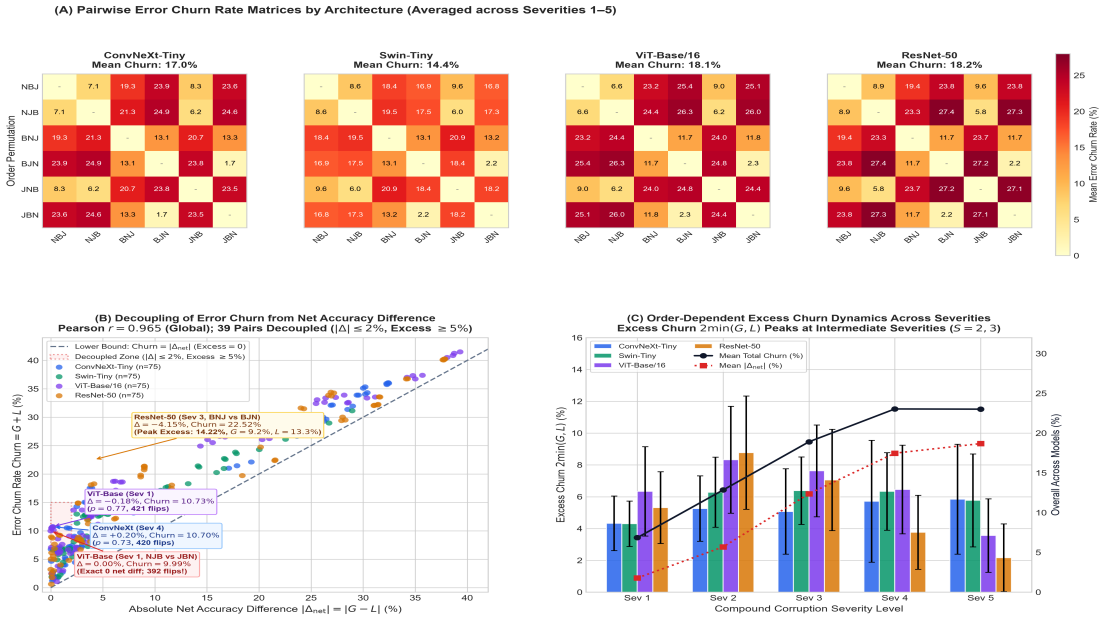


Figure 10: Order-Dependent Error Churn Analysis: (A) Pairwise churn rate matrices by architecture. (B) Decoupling of Error Churn from Net Difference (39 high-excess pairs). (C) Severity dynamics showing excess churn peak at intermediate levels.

Audited Insights on Prediction Churn: 1. In 39 of 300 sequence pairs, classifiers exhibit near-zero net accuracy shift ($|\Delta_{net}| \leq 2.0\%$) accompanied by large excess churn ($\geq 5.0\%$). In the extreme case of ViT-Base at Severity 1 (NJB vs JBN), aggregate accuracy is identically 73.17% ($\Delta = 0.00\%$, $p = 1.0$), yet 392 physical images flip correctness (Churn = 9.99%). 2. ViT-Base exhibits the highest mean excess churn (6.46%), reflecting global attention weight re-allocation under spatial phase shifts. 3. ResNet-50 exhibits peak excess churn of 14.22% (Sev 3, BNJ vs BJN, 360 rescues vs 524 breaks). 4. Confidence increases by ~+26% on Gain transitions and drops by ~-22% on Loss transitions.

9. Limitations & Honest Scope Assessment

To maintain scientific integrity for graduate admissions review, we explicitly articulate our scope limitations:

1. **10-Class Benchmark (Imagenette-v2):** Evaluations were conducted on 10 WordNet synsets (N=3,925 validation images) rather than the complete 1,000-class ImageNet-1K benchmark (N=50,000). Findings should not be extrapolated to fine-grained 1,000-class discrimination.
2. **Perturbation Diversity:** We evaluated 3 canonical perturbation families (15 shifted conditions in primary benchmark, 30 compound conditions in order study). We did not evaluate the full 19 corruption types of ImageNet-C or natural adversarial shifts (ImageNet-A, ImageNet-R).
3. **Pretraining Disparities:** Pretrained weights originate from standard open-source checkpoints: ConvNeXt-Tiny was pretrained on ImageNet-12K and fine-tuned on 1K; ViT-Base was pretrained on ImageNet-21K with AugReg and fine-tuned on 1K; ResNet-50 and Swin-Tiny were pretrained directly on ImageNet-1K. Pretraining scale is an unavoidable confounder.
4. **Task Specificity:** Limited strictly to 2D image classification; findings do not generalize directly to dense prediction (detection/segmentation).

10. Conclusion

RobustLens delivers an audited, reproducible benchmark investigating the robustness, uncertainty, and operator permutation sensitivity of modern vision architectures. By replacing naive un-clustered pooling with image-level clustered bootstrap confidence intervals and Monte Carlo paired sign-flip permutation tests, we resolve pseudoreplication. Our empirical results show that clean accuracy fails to predict out-of-distribution performance, that compound corruption order significantly alters accuracy and calibration, and that architectural inductive biases strongly mediate vulnerability to cascading degradation.

11. References

1. Angelopoulos, A. N., & Bates, S. (2021). A gentle introduction to conformal prediction. *arXiv:2107.07511*.
2. Bai, Y., et al. (2021). Are transformers more robust than CNNs? *NeurIPS*, 34, 26831–26843.
3. Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate. *JRSS-B*, 57(1), 289–300.
4. Bhojanapalli, S., et al. (2021). Understanding robustness of transformers for image classification. *ICCV*, 10231–10241.
5. D'Amour, A., et al. (2020). Underspecification presents challenges for credibility in modern machine learning. *JMLR*, 23(189), 1–61.
6. Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap Methods and their Application*. Cambridge University Press.
7. Deng, J., et al. (2009). ImageNet: A large-scale hierarchical image database. *CVPR*, 248–255.
8. Dosovitskiy, A., et al. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*.
9. Dünkel, P., et al. (2025). CNS-Bench: Benchmarking image classifier robustness under continuous nuisance shifts. *ICCV*.
10. Goyal, S., et al. (2022). Robust scene inference under noise-blur dual corruptions. *ICCP*, 1–11.
11. Guo, C., et al. (2017). On calibration of modern neural networks. *ICML*, PMLR, 1321–1330.
12. Hendrycks, D., & Dietterich, T. (2019). Benchmarking neural network robustness to common corruptions. *ICLR*.
13. Hendrycks, D., et al. (2021). Natural adversarial examples. *CVPR*, 15262–15271.
14. Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scand. J. Statist.*, 6(2), 65–70.
15. Howard, J. (2019). Imagenette: A smaller subset of 10 easily classified classes from ImageNet. *fastai/imagenette*.
16. Hurlbert, S. H. (1984). Pseudoreplication and the design of ecological field experiments. *Ecol. Monogr.*, 54(2), 187–211.
17. Lehmann, E. L., & Romano, J. P. (2005). *Testing Statistical Hypotheses* (3rd ed.). Springer.
18. Li, B., et al. (2025). LAION-C: An out-of-distribution benchmark for web-scale vision models. *ICML*.
19. Li, Y., et al. (2025). R-Bench: Are your large multimodal models robust to real-world corruptions? *IEEE JSTSP*.
20. Liu, Z., et al. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. *ICCV*, 10012–10022.
21. Liu, Z., et al. (2022). A ConvNet for the 2020s. *CVPR*, 11976–11986.
22. Michaelis, C., et al. (2019). Benchmarking robustness in object detection: Autonomous driving when winter is coming. *arXiv:1907.07484*.
23. Milani Fard, M., et al. (2016). Launch and iterate: Reducing prediction churn. *NeurIPS*, 29, 3179–3187.
24. Minderer, M., et al. (2021). Revisiting the calibration of modern neural networks. *NeurIPS*, 34, 15682–15694.
25. Mintun, E., et al. (2021). On interaction between augmentations and corruptions in ImageNet-C. *NeurIPS*, 34, 3571–3583.
26. Ovadia, Y., et al. (2019). Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. *NeurIPS*, 32.
27. Phipson, B., & Smyth, G. K. (2010). Permutation P-values should never be zero. *Stat. Appl. Genet. Mol. Biol.*, 9(1).
28. Recht, B., et al. (2019). Do ImageNet classifiers generalize to ImageNet? *ICML*, PMLR, 5389–5400.
29. Steiner, A., et al. (2022). How to train your ViT? Data, augmentation, and regularization in vision transformers. *TMLR*.
30. Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.